super keyword

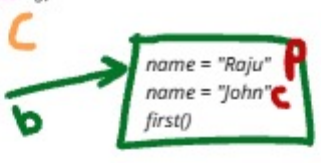
Super keyword is used to access the parent level information

1) In java methods participate in inheritance
2) In java constructors do no participate in inheritance
3) In java variables also participate in inheritance

=> if a methods is coming from parent class to the child class, then child has the permission to modify that method ==> Method Overriding

=> if a variable is coming from parent class to the child class, then child doesnt hav eth epower to modify the value of that variable

```
B b = new B();
```



Variable Shadowing

When child gets a method from parent, child can modify ==> M.O
When child gets a variable from a parent, child cannot modify ==> Variable Shadowing

when a child class gets a varoable from the aprent class, that variable will act like a shadow, and the child variable will act like a main person standing in the front of the shadow!

super keyword in java is used for accessing parent level information(parent class variables / parent class methods / parent class constructor)

```
class A
{
   String name = "Raju";

   public A()
   {
      System.out.println("I am from A class constructor!");
   }

   public void first()
   {
      System.out.println("Hello!");
   }
}

class B extends A
{
   String name = "John";

   @Override
   public void first()
   {
      System.out.println("Welcome");
   }
}


public class Second {

   public static void main(String[] args) {

      B b = new B();
      b.first();
      System.out.println(b.name);


   }

}
```

In java, if we dont have any constructor inside a class, then JAVA automaticallyu create a constructor ==> **DEFAULT CONSTRUCTOR**

Once the default constructir is created then insid ethata java is going to add just one line of code ==> super()

NOTE:-

to access parent class variable ==> super.variablename
to access parent class method ==> super.methodname
to access parent class constructor ==> super()

super keyword should always be used inside the child class only!

If a java class doesnt have any parent in teh code, then there is a defualt parent ==> Object class

```
class AA extends Object
{

   public AA() {
      super();
      System.out.println("AA constructor!");
   }

   String city = "Mumbai";

   public void first()
   {
      System.out.println("Welcome!");
   }
}

class BB extends AA
{
//   9 items + 3 items ==> 12 items ==> total methods ==> 9

   public BB()
   {
      super();
   }

   String city = "Banglaore";

   public void second()
   {
      System.out.println(city);
      System.out.println(super.city);
      super.first();
   }
}

   public class Third {
      public static void main(String[] args) {

         BB b = new BB();
         b.second();
         b.toString();


   //      System.out.println(super.city);
   //      System.out.println(super.city);

      }

   }
```

Explain type sof inheritance in java

1) Single Inheritance ==> This has exactly one child and for every one child we have exactly one parent

```
class AAA
{

}

class BBB extends AAA
{

}
```

2) multi level inheritance

```
//multilevel inheritance

class AAA extends Object
{

}

class BBB extends AAA
{

}

   class CCC extends BBB
   {

   }
```

3) heirarchial inheritance ==> same parent can be there for multipel childeren

```
//multilevel inheritance

class AAA extends Object
{

}

class BBB extends AAA
{

}

class CCC extends AAA
{

}
```

4) Multiple Inheritance ==> multiple parents for single child

Multiple Inheritance is not possible in java using classes but it is posisble using interface

```
//multilevel inheritance

class AAA
{
   public void first()
   {

   }
}

class BBB
{
   public void first()
   {

   }
}

class CCC extends AAA, BBB
{
//   first() and first()
}
```